Categorical Problems 2 & 3

Saturday, December 9, 2017 11:04 AM

$$y = \alpha + \beta_1 \cdot x + \beta_2 \cdot x \cdot 1_S + \beta_3 \cdot x 1_R$$

$$y = 2 + 6 \cdot x - 2 \cdot x \cdot 1_S - 1 \cdot x \cdot 1_R$$

Sunny: $y = 2 + 6 \cdot x - 2 \cdot x \overset{1}{1_S} - \cancel{1 \cdot x \cdot 1_R}^{0}$

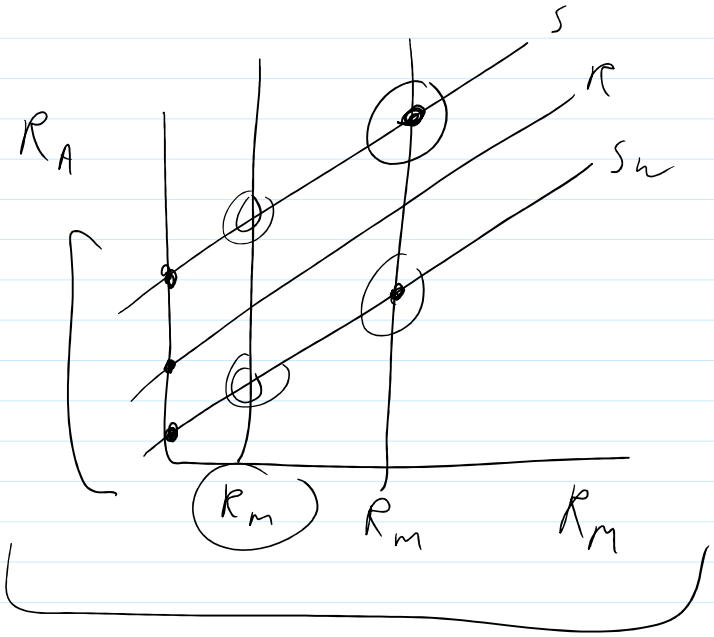
$= 2 + 6 \cdot x - 2 \cdot x$

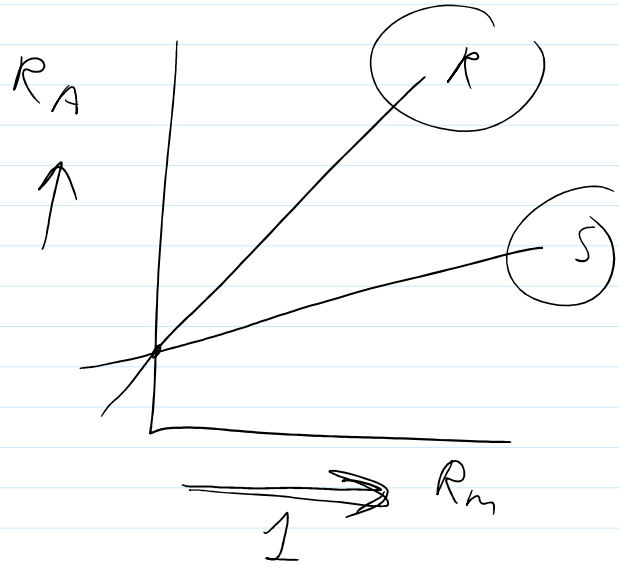
$y = 2 + (6-2)x = 2 + 4x$

Rainy: $y = 2 + 6 \cdot x - \cancel{2 x 1_S}^{0} - 1 \cdot x \overset{1}{1_R}$

$y = 2 + 6x - 1x = 2 + 5x$

Snow: $\boxed{y = 2 + 6x} - \cancel{2 x 1_S}^{0} - \cancel{1 x 1_R}^{0}$





$$R_A = \alpha + \beta_1 \cdot R_m + \beta_2 \cdot 1_S + \beta_3 \cdot 1_R$$

$$= 1.9 + 5.4 \times R_m - 0.4\, 1_S + 0.2 \cdot 1_R$$

Sunny: $R_A = 1.9 + 5.4 \times R_m - 0.4 \cdot \overset{2}{1_S} + \cancel{0.2 \cdot \overset{0}{1_R}}$

$$= (1.9 - 0.4) + 5.4 \times R_m$$

$$R_A = 1.5 + 5.4 \times R_m$$

Rainy: $R_A = 1.9 + 5.4 \cdot R_m \cancel{- 0.4 \cdot \overset{0}{1_S}} + 0.2 \cdot \overset{2}{1_R}$

$$= (1.9 + 0.2) + 5.4\, R_m$$

$$R_A = 2.1 + 5.4\, R_m$$

Snow: $\boxed{R_A = 1.9 + 5.4\, R_m}$ $\cancel{.4 \overset{0}{1_S} + .2 \times \overset{0}{1_R}}$